

Chapter 1 — Personal and Professional Development of the Urologist

Part I: Clinical Decision Making

- Earlier editions addressed knowledge and skills; **professional attitudes and attributes** are now treated as core content.
- **Professional formation is ancient:** Hippocrates taught his students ethics, duty, obligation and personal decorum, including dress and behaviour.
- **Breach of professionalism as a medical student is one of the most predictive factors for later disciplinary action** by medical boards; state and medical society boards recognise professionalism as essential.
- **Development is lifelong** — not complete at the end of medical school or residency; it needs ongoing reflection.
- **Mental and physical health are inseparable**, and the physician's own health underpins professional conduct and patient care; burnout degrades performance and patient outcomes.
- **Ultimate goal:** the health and longevity of the physician and of the physician's patients.

PERSONAL DEVELOPMENT

- **Stimulus → reflex reaction → chosen response:** the initial reaction may be hard-wired, but the subsequent response is under personal control; this capacity allows one to direct personal development.
- **Growth towards goals aligned with life purpose** gives reward, joy and energy.
- **Method: define core values and goals → prioritise activities that serve them → build routines and habits → review periodically.**
- **Learning from adversity** — drawing lessons from challenge, stress or suffering shifts perspective towards a "silver lining".
- **Reflection on professional ethics and career goals begins in medical school** and is deliberately re-examined, refined and refreshed.
- **Disruptive events are inevitable** ("Everyone has a plan until they get punched in the mouth" — Mike Tyson) and force reassessment of priorities, goals and even identity; **a clear prior understanding of one's beliefs, values and core principles** allows new plans congruent with personal and professional ethics.

Career Development and the Life Cycle of a Urologist

- **The career cycle is finite and nested within the longer human life cycle;** accepting this early aids planning of career development and transitions.
- **Cellular model of life (Fig. 1.1):** life as a cell, its cycles as organelles. Each cycle's products affect the others and the whole cell; activity varies with need and external factors.

FIG. 1.1 CELLULAR MODEL OF LIFE — THE CYCLES

Cycle	Components shown
Career	The urology career cycle
Physical	Physical health
Mental	Soul, spirit, cognition, creation, identity
Social	Family and friends, colleagues

PHASES OF THE UROLOGY CAREER CYCLE

Phase	Timing	Priorities	Hazard
Conception	Medical school	Reflection on ethics and goals begins	—
Embryological	Residency	Training	—
Early career	About 10–12 years	Say yes: set goals and plans; find advisers, mentors, coaches; build reputation; start long-term financial plan; climb learning curves (difficult patients, clinical situations, procedures), consulting senior urologists	Excessive workload; burnout risk may be highest
Mid career	Mid-40s to mid-50s; about 10–12 years	Peak surgical performance (experience, technique, stamina); increasing scrutiny — weigh each opportunity's importance against time and effort	Burnout common; selective acceptance is protective
Late career	About 10–12 years	Decreasing clinical activity , offset by administration, leadership, teaching, mentoring, coaching; develop social activities and hobbies	—
Retirement	End of career cycle; outlived by the life cycle	Transition planning	Struggle if career and life cycles were falsely equated: loss of identity, belonging and purpose

- **Mentorship at every stage:** others have travelled the path; most will serve as counsellors, mentors or coaches if asked. **Peer groups serving these roles for one another** add collegiality and guard against isolation and burnout.
- **First impressions last:** accepting and delivering on early opportunities generates further offers (referrals, research, teaching, administration, leadership, business) and makes staff and colleagues want to work with the urologist.
- **Time and effort are finite:** overinvestment in the career cycle starves the other cycles and harms the whole.

Concept — *The career is one finite organelle within the cell of life; every phase-specific priority follows from protecting the whole cell rather than maximising the career cycle.*

Burnout

- **Definition:** a symptom constellation of **emotional and ethical exhaustion, depersonalisation and diminished personal accomplishment at work**, in response to chronic emotional and interpersonal stressors.
- **A distinct entity** from job dissatisfaction, fatigue, occupational stress and depression, although associated with each.
- **Prevalence:** around 40% of United States physicians, including trainees, for several decades; recent figures approach 50% in urologists.
- **Rising despite attention:** American Urological Association (AUA) census 2021 versus 2016 showed increased burnout among urologists, notably in women and in those under 45. Mid-career physicians, historically the highest group, fell slightly below early-career urologists.
- **Women have higher burnout**, and female urologists are more often dissatisfied with work–life balance and more often report too little time for personal or family life.

- **Physician comorbidities:** heart disease, diabetes, insomnia, anxiety, substance abuse, depression and suicide.
- **Physician suicide:** about one United States physician per day; rate relative to the general population **1.4 times in men and 2.7 times in women; 12–14% of burned-out physicians have suicidal ideation.** Help-seeking is impeded by **fear of losing the medical licence.** Resources: physiciansupportline.com; 988lifeline.org.
- **Effects on patients and workforce:** lower patient satisfaction and safety, lower-quality care, more medical error and poor clinical judgement; lower productivity and job satisfaction; physicians leave practice → workforce shortage → delayed urological care.
- **Contributors:** loss of autonomy; loss of purpose (time on tasks not requiring a physician's training); loss of community and isolation from colleagues; electronic health record (EHR) time; excessive workload; insufficient personal and family time; education debt; falling reimbursement; rising regulation and metrics; administrative burden.
- **Moral injury:** when these factors prevent the physician from giving the quality of care they wish, the result is stress and moral injury that feeds burnout.
- **The 20% threshold:** physicians spending **less than 20% of work effort on the activity they find most meaningful** are almost three times as likely to be burned out.

BURNOUT PREVENTION STRATEGIES

Strategy	Examples	Effect
Institutional seminars	Awareness sessions	Minimally effective for prevention
Direct practice interventions	Scribes ; removing daily work frustrations; efficiency	More effective
Restoring autonomy	Influence over work environment and schedule	Significantly improves wellness , reduces burnout
Department-wide community programmes	Weekly lunches, wellness days, holiday gatherings	Reduced burnout scores; staff feel appreciated and valued
Belonging and connection	Mentorship, coaching, small peer groups, social events	Recently advocated; logical for belonging and meaning
Individual, evidence-based	Mindfulness-based stress reduction, communication skills training, exercise, self-care programmes	Evidence-based reduction

Concept — *Burnout is driven by **loss of autonomy, purpose and community**; interventions that restore one of these work, whereas awareness alone does not.*

Maintaining Morale

- **Anchor on core goals and values:** career goals serve overarching personal goals and purpose; routine review keeps sight of "the forest for the trees" and allows course correction. Effort towards core goals is energising.
- **Service to others is the value common to all physicians.** Among peers, service seems commonplace; remember how selected, trained, needed and respected each physician is. A "routine" visit may be the most important part of a patient's day or week, and small positive changes can cascade into major impact.
- **The physician as role model** to staff, students, colleagues, patients, families and society: care for patients demonstrates the inherent value and sanctity of every person; scientifically grounded counsel models thoughtful responsiveness. Recalling this impact restores self-image and self-esteem.

- **One's own role models:** imitating inspiring colleagues and teachers sustains progress and perspective in difficult times; remember that one is also being watched.
- **Service itself is the reward** — an honour and a privilege. Research, teaching, mentorship and administration are alternative routes to service, giving flexibility.
- **Expect adjustment:** reflection may refine or discard goals, even to a change of career; **career goals are subservient to personal growth goals.**

Concept — *Morale is sustained by reconnecting daily work to the purpose of serving others; when the work no longer serves that purpose, the work is changed, not the purpose.*

Sustainable Change and Self-Care

- **Change is inevitable**, in both the internal and external (work) environment. Drivers: exponential growth of technology and scientific knowledge in the 20th and 21st centuries and **corporatisation of United States health care from the 1960s.**
- Hence **lifelong learning was built into the American Board of Urology maintenance of certification** programme.
- **Change is beyond control; the response is not.** After the initial reaction, respond by adjusting the cycles of Fig. 1.1 — temporarily or for longer: meditation, physical activity, work activity, social support, a new skill — directed at happiness, security, purpose or future opportunity.
- **Build the non-career cycles before the crisis** — "broadening the portfolio": investments in the mental, physical and social cycles pay dividends to the other cycles and give more response options in hard times.
- **Habits** are a powerful determinant of behaviour, often unrecognised without reflection, and can be deliberately constructed; **an activity becomes an established habit after about 3 weeks.**
- **Subtraction as well as addition:** a response cannot always be a net addition; identify activities that no longer give energy or growth and modify or drop them.

Concept — *Resilience is pre-built capacity: the options available in a crisis are only those already developed in the physical, mental and social cycles.*

Complications and Shortcomings

- **All physicians make mistakes and have complications** — "the only way to avoid operative complications is to stop operating".
- **Systems view:** many complications arise from a **combination of potentially avoidable events within a system**, not a single individual failure.
- **Reframe failure as an opportunity to learn and grow;** recognising suffering as part of the shared human condition reduces feelings of being overwhelmed and isolated.
- **Self-compassion** — how we relate to ourselves in perceived failure or inadequacy — is **learnable**; "To err is human, to forgive is divine" (Alexander Pope) applies to oneself.
- Self-compassion is associated with **more** mindfulness, happiness, life satisfaction, resilience and health-promoting behaviour, and **less** shame, depression, anxiety, stress and suicidal ideation.

Concept — *Complications are system events and learning opportunities; self-compassion rather than self-blame converts them into growth.*

DEVELOPMENT OF PROFESSIONAL SKILLS, KNOWLEDGE AND BEHAVIOURS

Core Attributes

- **Shared with procedural specialties:** leadership, decision-making, problem-solving and analytical skills.

- **Characteristic of urologists:** excellent visual-spatial psychomotor ability, strong sense of autonomy, sense of humour, keen interest in pressure-flow dynamics.
- **Commitment to lifelong learning** is particularly important given urology's affinity for new skills and surgical technology.

The Science of Learning

THE FOUR DOMAINS OF LEARNING

Domain	Content	Framework	Acquisition and assessment
Knowledge	Facts to higher-order cognition	Bloom's taxonomy (Fig. 1.2)	Easiest to test — certification exams rely heavily on it
Psychomotor	Technical skills	Stages of technical skills acquisition (Fig. 1.3)	Courses, proctorship, fellowships, simulation
Communication	Including professionalism and interpersonal skills	TeamSTEPPS (Fig. 1.4)	Formal residency curriculum often lacking
Affective	Feelings, values, emotions, motivations, acceptance, attitudes, attentional abilities	Krathwohl's taxonomy	Increasingly in formative assessment

Knowledge Domain

FIG. 1.2 BLOOM'S TAXONOMY — COGNITIVE DOMAIN (2001)

Order	Level (lowest to highest)	Meaning
Lower-order	Remembering	Recall specific facts
Lower-order	Understanding	Grasp meaning of instructional materials
Lower-order	Applying	Use information in a new but similar situation
Higher-order	Analysing	Take apart the known and identify relationships
Higher-order	Evaluating	Examine information and make judgements
Higher-order	Creating (apex)	Use information to create something new

- **Sources:** comprehensive textbooks (Campbell's Urology, AUA Core Curriculum); for currency — journals, AUA University, online videos and blogs, courses, proctorships, meetings, observerships.
- **Early habits of continuous learning are an ethical and professional obligation** — for maintenance of competence and the path to expertise and mastery.

Psychomotor Domain

- **Technology-driven field:** urologists must maintain skills and **periodically retool** as devices and techniques arrive; infrastructure must ensure safe transition of innovation into patient care.
- **Proctorship is often required for credentialing** at local hospitals in specialty skills such as **robotic and laser surgery**.
- **Routes:** specialty- and industry-sponsored hands-on courses, proctorships, exchange programmes, formal fellowships, mini-fellowships, simulation.
- **Simulation** serves to acquire and maintain skills and to **warm up before operating**. Models of varying validity: synthetic part-task trainers and complete surgical systems, virtual reality trainers, biological tissue, live animals.

FIG. 1.3 UNDERSTANDING TECHNICAL SKILLS ACQUISITION

Stage	Level	Descriptor	Recollection	Recognition	Decision	Awareness
Imitation	Novice	Observes and patterns behaviour after someone else	Non-situational	Decomposed	Analytical	Monitoring
Manipulation	Competent	Performs after training, with errors	Situational	Decomposed	Analytical	Monitoring
Precision	Proficient	Refines technique, few errors	Situational	Holistic	Analytical	Monitoring
Articulation	Expert	Able to multitask	Situational	Holistic	Intuitive	Monitoring
Naturalisation	Master	High-level performance on autopilot	Situational	Holistic	Intuitive	Absorbed

- **Pattern:** each stage converts one element of mental functioning, in order — recollection (competent), recognition (proficient), decision (expert), awareness (master).
- **Trap:** the expert is intuitive but still monitoring; only the master is absorbed.

Communication Skills Domain

- **Paramount** with patients, colleagues and staff: routine interactions plus advanced skills such as **breaking bad news and de-escalation**.
- **Communication errors contribute to about 70% of medical errors and 27% of malpractice claims.** Poor communication also lowers treatment adherence and patient satisfaction and wastes resources.
- **A formal communication curriculum is often lacking in urology residency.**
- **TeamSTEPPS (Team Strategies and Tools to Enhance Performance and Patient Safety)**, supported by the American Hospital Association (AHA): a framework for formal training in practices and institutions that builds a culture of safety. Derived from **high-reliability principles** and more than 30 years of research; aims — teamwork, communication, every member able to speak up, employee satisfaction.
- **Evidence:** improves attitudes, knowledge and behavioural skills; **combined with simulation, in some cases improves patient outcomes.** A meta-analysis of teamwork training showed a positive effect on team performance and patient outcomes; training should cover **both routine and non-routine situations**.
- **Train individuals to be good team members** when whole-team training is impractical; the physician as leader sets the culture and tone. Local and national courses are available through the AHA.

FIG. 1.4 TEAMSTEPPS CONCEPTS

Concept	Definition
Situation monitoring	Actively scanning and assessing elements of the situation to gain information or maintain accurate understanding of the team's situation — a continuous process
Situation awareness	Product of situation monitoring — individual outcome
Shared mental model	Product of situation monitoring across members — team outcome
Brief	Short planning session before the start
Huddle	Team regroup to re-establish awareness and planning
Debrief	Information exchange after the action
Call-out	To request or provide information
Check-back; cross-check	Repeating back ("parroting") to confirm understanding — closing the loop of communication
SBAR	Situation, Background, Assessment, Recommendation
Hand-off	Transfer of information during transitions

Concept	Definition
CUS	I'm Concerned, I'm Uncomfortable, this is a Safety issue
Two-challenge	Responsibility to assertively voice a concern at least twice to ensure it has been heard

Concept — *Because communication failure is the dominant contributor to medical error, communication is trained as a structured team skill with named tools, not left to personality.*

Affective Domain

KRATHWOHL'S TAXONOMY OF THE AFFECTIVE DOMAIN

Level (ascending)	Meaning	Examples
1. Receiving	Willingness to learn	Listening open-mindedly to controversial issues; respecting others' rights; remembering names of newly introduced people
2. Responding	Levels of active participation	Doing assigned reading; preparing for cases; questioning new ideas; independent research
3. Valuing	Worth attached to a phenomenon or behaviour	Interest in good patient outcomes; active problem-solving; speaking up about a threat to patient care
4. Organisation	Bringing values together and resolving conflicts into an internally consistent value system	Recognising own abilities, limitations and values; realistic aspirations; accepting responsibility for own behaviour and professional ethical and time-management standards — very important for a safe, effective urologist
5. Characterisation by a value set	Patterns of adjustment developed over time	Self-reliance working independently; objective problem-solving; commitment to ethical practice; revising judgement in light of new evidence

- **Society of Academic Urologists Curriculum Task Force:** training programmes now include affective and communication elements in **formative assessment**.

Concept — *Certification tests the domain easiest to assess (knowledge), whereas error and harm arise largely in the harder-to-assess communication, affective and psychomotor domains — hence the need for deliberate training in them.*

Responsibility for Development and Maintenance of Skills

WHO ENSURES A UROLOGIST'S COMPETENCE

Party	Role
Public	Has a stake in the urologist's skills
Hospitals	Credentialing ; share a large part of the risk
Regional licensing and specialty boards	Policing to protect the public from incompetent or dangerous urologists; specialty boards assess mainly the knowledge domain by objective standardised assessment and struggle with the other domains
Peer review	Critical for the non-knowledge domains; the American Board of Urology uses surgical logs to assess outcomes and indications as a surrogate for technical assessment
Payers	Incentive to ensure competence; control network membership

Party	Role
Individual urologist	Ultimate responsibility across all domains, guided by a "vision of being better" and innovation anchored in ethical and moral values

Concept — *Many parties police competence, but each sees only part of it; only the individual urologist is accountable across all four domains.*

ORGANISATIONAL SUPPORT OF PROFESSIONALISM

- **Regulators and certification boards** (state licensure boards, The Joint Commission) take an interest in provider professionalism.
- **Domains of professionalism:** availability, optimal communication, team-based contribution, respect, technical and cognitive competence, self-awareness — affecting colleagues, co-workers, patients and families.
- **Promotion requires contributions from the individual, group, institution and care system.**
- **Elements of an optimal culture of safety:** a supportive culture of service; **oversight** (identifying variance from optimal professional activity, data collection and aggregation, feedback to individuals and groups); support systems (wellness and employee assistance programmes); leadership support. The result is a **high-reliability organisation** with less unnecessary variation in care and human-interaction outcomes.
- **Infrastructure:** committed leaders and dedicated teams; organisational cultural goals and values; **enforced policies and procedures**; resources for engaged teams; **a tiered intervention model**; data reporting, assessment and reproducible review; institutional support for education and training.

PATIENT VERSUS COWORKER REPORTING SYSTEMS

Feature	Patient Advocacy Reporting System (PARS)	Coworker Reporting System (CORS)
Source of observations	Patients and families	Co-workers
Interaction modulated	Provider–patient/family	Provider–team
Delivery	Trained peer messengers in both	
Demonstrated effect	Substantially improved patient experience; reduced professional and institutional malpractice exposure	Robust behaviour change — most reported individuals are never reported again ; improves culture of safety; manages disrupted micro-environments in large institutions
Common use	Tracking and trending of individual providers; directs specific interventions — behavioural intervention, anger management, wellness activities	

- **Surgical outcome signal:** patients whose surgeon is in the **highest quartile of unsolicited patient observations** have a higher risk of surgical complications than those whose surgeon is in the lowest quartile.
- Supporting the individual practitioner and their group interactions underpins organisational culture, individual wellness and team function — part of continuous improvement of care delivery and culture.

Concept — **Unprofessional behaviour is a measurable safety signal:** *complaints predict complications, so organisations collect, trend and feed back behaviour through tiered peer-messenger systems rather than waiting for harm.*